Spectral Statistics of Deterministic Arithmetic Prime Matrices: A Phase Diagram from GOE to Poisson

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Abstract

We systematically study the spectral statistics of deterministic matrices defined by arithmetic functions on prime sums, mapping five kernels—sinc, divisor d, Liouville λ, Möbius μ, and von Mangoldt Λ—in the parameter space spanned by matrix density, sign structure, and value discreteness. Using the unfolding-free Oganesyan–Huse spacing ratio $\langle\tilde{r}\rangle$ and a battery of control experiments, we establish a continuous spectrum from the Gaussian Orthogonal Ensemble (GOE) through Poisson to a sub-Poisson compressed regime.

Our main findings are: (1) dense kernels $d(p_i+p_j)$ and $\lambda(p_i+p_j)$ exhibit GOE statistics ($\langle\tilde{r}\rangle \approx 0.53$), providing numerical support for an additive version of the Chowla conjecture; (2) sparsity is the dominant tuning parameter, with $\langle\tilde{r}\rangle$ decreasing monotonically from GOE ($\rho = 1$) to a compressed state ($\rho = 0.0008$); (3) at fixed extreme sparsity, the discrete value set of Λ contributes an additional compression beyond the Poisson baseline ($\Delta\langle\tilde{r}\rangle \approx 0.08$, 95% significant); (4) sign structure (non-negative vs. alternating) has no significant independent effect at any tested density; (5) none of the tested deterministic constructions produce GUE statistics. This phase diagram provides a systematic empirical classification of spectral statistics for deterministic arithmetic matrices and delineates the current boundary of what is achievable without injecting independent randomness.

1 Introduction

In preceding work [1, 2], we established a rigorous spectral theory for the sinc-kernel Gram matrix on the primes [1] and studied non-convolution arithmetic matrices based on the von Mangoldt function [2]. The sinc kernel produces GOE statistics with effective rank $\Theta(\log N)$, while the extremely sparse Λ-kernel ($\rho \approx 0.0008$) produces a sub-Poisson compressed state with $\langle\tilde{r}\rangle \approx 0.31$. These findings raise two fundamental questions:

(1) Is the transition from GOE to the compressed state continuous or discontinuous? What is the tuning parameter?

(2) Can any deterministic arithmetic construction produce GUE statistics?

We answer these questions by systematically mapping five arithmetic kernels in the parameter space of *density* (non-zero fraction), *sign structure* (non-negative / alternating), and *value discreteness* (continuous / discrete). All matrices are defined on the first $N = 2000$ primes with zero diagonal (except the sinc kernel, whose diagonal is 1).

Kernel	K_{ij} $(i \neq j)$	Density	Sign	Values
sinc	$\mathrm{sinc}(\pi \log(p_i/p_j))$	100%	mixed	cont. $[-1, 1]$
d	$d(p_i+p_j)$	100%	≥ 0	disc. $2, 3, 4, \ldots$
λ	$\lambda(p_i+p_j)$	100%	± 1	disc. $\{-1, 1\}$
μ	$\mu(p_i+p_j)$	37%	$\{-1, 0, 1\}$	disc. $\{-1, 0, 1\}$
Λ	$\Lambda(p_i+p_j)/\log(p_i+p_j)$	0.08%	≥ 0	disc. $1, \frac{1}{2}, \frac{1}{3}, \ldots$

Table 1: The five arithmetic kernels. $d(n)$ is the divisor function, $\lambda(n) = (-1)^{\Omega(n)}$ the Liouville function, $\mu(n)$ the Möbius function, and $\Lambda(n)$ the von Mangoldt function.

2 Kernel definitions and experimental design

2.1 Five arithmetic kernels

2.2 Control experiments

For sparse kernels (μ, Λ), we run all five controls: GOE dense, random sparse (density-matched), structured sparse (same positions, random values), NonNegSparse (same positions, Exp(1) values), and value permutation (same positions and value set, shuffled). For dense kernels (d, λ), sparsity-related controls do not apply; we run NonNegSparse, value permutation, and sign randomization (for λ).

2.3 Statistic

We use the unfolding-free Oganesyan–Huse spacing ratio [3]:

$$\tilde{r}_k = \frac{\min(s_k, s_{k+1})}{\max(s_k, s_{k+1})}, \qquad s_k = \lambda_{k+1} - \lambda_k. \tag{1}$$

Reference values: Poisson ≈ 0.386, GOE ≈ 0.536, GUE ≈ 0.602.

3 Results

3.1 Spacing ratio means

Kernel	Density	Sign	Pos. eig.	$\langle \tilde{r} \rangle$	Class
d	100%	≥ 0	468	0.529 ± 0.012	GOE
λ	100%	± 1	501	0.530 ± 0.011	GOE
μ	37%	$\pm 1, 0$	501	0.495 ± 0.012	transitional
Λ	0.08%	≥ 0	292	0.309 ± 0.031	compressed

Table 2: Spacing ratio statistics at $N = 2000$. The sinc kernel is excluded: its $\Theta(\log N)$ effective rank yields too few eigenvalues for reliable ratio statistics; its GOE classification comes from the unfolding analysis in [1]. Dense kernels (d, λ) have ~ 500 positive eigenvalues; the Λ-kernel has 292.

3.2 Effect decomposition

3.3 Key findings

Finding 3.1 (Dense-kernel GOE universality)**.** Both $d(p_i+p_j)$ and $\lambda(p_i+p_j)$ produce GOE statistics at 100% density, within one standard error of the theoretical value 0.536. Value

Kernel	Control	$\langle \tilde{r} \rangle$	p	Sig.
d (0.529)	NonNegSparse	0.527	0.86	ns
	Permutation	0.530	0.96	ns
λ (0.530)	NonNegSparse	0.533	0.72	ns
	Permutation	0.528	0.84	ns
	SignRandom	0.531	0.95	ns
μ (0.495)	SparseRandom	0.498	0.86	ns
	NonNegSparse	0.493	0.91	ns
Λ (0.309)	NonNegSparse	0.388	0.013	**
	Permutation	0.279	0.34	ns

Table 3: Control experiments. "Sig." indicates significance at 95%. Only Λ vs. NonNegSparse is significant; all other differences are within statistical noise.

permutation and sign randomization controls confirm that neither the spatial arrangement of values nor the sign structure (+ vs. $\pm$) has any significant effect.

Finding 3.2 (Sparsity as dominant tuning parameter)**.** $\langle \tilde{r} \rangle$ decreases monotonically with decreasing density: 100% $\to$ 0.53 (GOE), 37% $\to$ 0.50 (transitional), 0.08% $\to$ 0.39 (Poisson baseline from NonNegSparse).

Finding 3.3 (Λ-kernel discrete-value compression)**.** At fixed extreme sparsity (0.08%), the Λ-kernel's $\langle \tilde{r} \rangle = 0.309$ is significantly below the NonNegSparse baseline (0.388, $p = 0.013$, Welch $t = -2.56$, $\nu \approx 68$), confirming the discrete-value-set effect established in [2].

Finding 3.4 (Sign structure is irrelevant at all densities)**.** At 100% density, d (all positive) and λ (alternating ± 1) give statistically indistinguishable $\langle \tilde{r} \rangle$ ($p \approx 0.9$). At 37% density, μ vs. NonNegSparse is also insignificant ($p = 0.91$).

Finding 3.5 (GUE is not reached)**.** None of the five deterministic kernels exceed $\langle \tilde{r} \rangle = 0.536$. GUE ($\approx 0.602$) appears only in random perturbation controls [1].

3.4 Positive eigenvalue count

Dense kernels (d, λ) and the medium-sparse kernel (μ) all yield positive eigenvalue fractions of 23–25%, while the extremely sparse Λ-kernel yields only 7.8%. Both μ and λ give 501 positive eigenvalues, consistent with their value-domain means being zero ($\mu \in \{-1, 0, 1\}$, $\lambda \in \{-1, 1\}$), producing approximately symmetric positive/negative spectra. The non-negative kernels (d, Λ) have lower positive fractions due to their one-sided value distributions.

3.5 Global spectral moments

To complement the local spacing statistics, we compute the scale-invariant moment ratio

$$\kappa_N = \frac{N \cdot \mathrm{Tr}(K^4)}{(\mathrm{Tr}(K^2))^2}, \tag{2}$$

whose theoretical value for the semicircle law (shared by GOE, GUE, and GSE) is $\kappa = 2$.
Finding 6 (λ: full GOE confirmation). The λ-kernel is the unique tested construction that simultaneously passes both global ($\kappa_N \to 2$) and local ($\langle \tilde{r} \rangle \approx 0.53$) GOE tests. This constitutes the strongest numerical evidence for an additive Chowla-type statement.
Finding 7 (d, Λ: global non-semicircle). The d and Λ kernels have $\kappa_N \to \infty$ because their matrix entries grow with N ($d(n) \sim \log n$, Λ values fixed but sparse). Their global spectral

Kernel	N=500	N=1000	N=2000	Limit
λ	1.936	1.966	1.985	$\to 2$
μ	2.072	2.027	2.007	$\to 2$
d	270	498	918	$\to \infty$
Λ	243	489	982	$\to \infty$
Semicircle	2	2	2	2

Table 4: Scale-invariant moment ratio κ_N. The λ and μ kernels converge to the semicircle value 2; d and Λ diverge due to unbounded matrix entries.

distributions have heavy tails, not semicircular. The d kernel nonetheless exhibits local GOE ($\langle \tilde{r} \rangle \approx 0.53$), demonstrating that local and global statistics can diverge.
Finding 8 (μ: global semicircle, local transitional). The μ-kernel achieves $\kappa_N \approx 2.007$ at $N = 2000$, matching the semicircle law to within 0.4%. Its local statistic $\langle \tilde{r} \rangle = 0.496$ is transitional between Poisson and GOE.

3.6 Bulk spectrum analysis

For d and Λ, we recomputed κ_N after removing the largest eigenvalue (Perron–Frobenius dominant eigenvalue). For d: κ increased from 918 to 1069, confirming that the divergence is a *bulk* property (the entry-growth effect) rather than an outlier artifact. For Λ: κ increased from 982 to 1928, confirming that the dominant eigenvalue was *suppressing* the divergence, not causing it.

4 Discussion

4.1 Connection to the Chowla conjecture

The GOE statistics of $\lambda(p_i+p_j)$ indicate that the Liouville function's values on prime sums possess sufficient pseudo-randomness to activate Wigner–Dyson universality. This is consistent in spirit with the Chowla conjecture [4], which asserts the non-correlation of λ on consecutive integers. Tao [5] established logarithmic averaged versions of the Chowla conjecture for two-point correlations. Our observation that the full matrix $\lambda(p_i+p_j)$ exhibits GOE statistics provides numerical evidence for a stronger, matrix-level statement. Rigorously establishing the connection between the weak correlation of matrix entries and the universality of spectral statistics—i.e., verifying the conditions of Wigner–Dyson universality theorems for this specific matrix ensemble—is an open problem.

4.2 Sparsity scaling law

The data suggest that $\langle \tilde{r} \rangle$ increases monotonically with density ρ, approaching GOE (≈ 0.536) as $\rho \to 1$ and Poisson (≈ 0.386) as $\rho \to 0$. The Λ-kernel's discrete-value effect ($\Delta\langle \tilde{r} \rangle \approx 0.08$) constitutes a correction term superimposed on the sparsity baseline at $\rho \ll 1$. Mapping more kernels at intermediate densities $\rho \in (0.01, 0.5)$ is needed to determine the precise functional form of this scaling law.

4.3 Implications for the GUE boundary

Our data strongly support the following empirical conjecture: within the framework of finite-dimensional, explicitly constructed, real symmetric matrices on prime indices, GUE is unreachable. This is not a claim about all deterministic matrices—known examples such as circulant matrices, Toeplitz matrices, and Laplacians on arithmetic hyperbolic manifolds can produce non-GOE statistics—but rather an empirical boundary for our specific construction family. A rigorous proof would require combining random matrix universality theorems with correlation decay estimates from additive number theory.

5 Conclusion

We have mapped the spectral statistics of five deterministic arithmetic prime kernels in the density–sign–discreteness parameter space. Dense kernels universally exhibit GOE; sparsity is the dominant parameter governing the GOE-to-Poisson transition; the Λ-kernel's discrete value set contributes additional compression at extreme sparsity; sign structure has no independent effect at any density; and GUE remains unreachable by any tested construction. This phase diagram provides a systematic empirical foundation for the spectral classification of deterministic arithmetic matrices and delineates the current boundary of what is achievable without independent randomness.

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References

[1] [Anonymous], *On the Spectral Structure of Sinc-Kernel Matrices on the Primes: Effective Rank, Geometric Decay, and Frequency-Box Decomposition*, 2026.

[2] [Anonymous], *Non-Convolution Arithmetic Matrices on the Primes: Breaking the Effective-Rank Barrier, Effect Decomposition, and Non-Universal Spectral Statistics*, 2026.

[3] V. Oganesyan and D. A. Huse, Localization of interacting fermions at high temperature, *Phys. Rev. B* **75**, 155111, 2007.

[4] S. Chowla, *The Riemann Hypothesis and Hilbert's Tenth Problem*, Gordon and Breach, 1965.

[5] T. Tao, The logarithmically averaged Chowla and Elliott conjectures for two-point correlations, *Forum of Mathematics, Pi* **6**, e8, 2018.